



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	INDUSTRIAL QUALITY MANAGEMENT				
Type of Course	VOCATIONAL MINOR				
Semester	VIII				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Fundamentals of Quality Management: Familiarity with basic quality management concepts, principles, and methodologies can be beneficial. This includes knowledge of quality standards (e.g., ISO 9001), quality improvement techniques (e.g., Six Sigma, Lean), and quality assurance processes. 2. Problem-Solving Skills: Strong problem-solving and critical thinking skills are crucial for identifying quality issues, analyzing root causes, and developing effective solutions.				
Course Summary	To give the students a thorough knowledge about the characteristics, functions and principles of management, administration and management, strategic management process, quality management assistance tools, total quality management tools & techniques, human resource development and management information systems.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Foundational understanding of management principles, functions, theories, and organizational structures, enabling the students to apply this knowledge in practical contexts within various types of organizations.	U	C	Instructor-created exams / Quiz /Assignment
CO2	Students equipped with the knowledge and skills to analyze business environments, formulate effective strategies, evaluate strategic options, and implement strategic plans to achieve organizational success and competitive advantage.	An	C	Class test /Assignment /Quiz

CO3	Able the students to apply quality management principles, tools, and standards to improve processes, enhance product/service quality, meet customer expectations, and ensure organizational excellence.	Ap	P	Assignment/ Class test/Seminar presentation
CO4	Enable the students to apply TQM tools and techniques to improve organizational processes, enhance teamwork, foster customer-centric approaches, and understand the strategic role of HRM in achieving organizational goals and objectives.	Ap	P	Assignments /Seminar presentation
CO5	Have a solid understanding of Management Information Systems, their components, functions, and strategic importance in supporting decision-making, enhancing business processes, and enabling digital transformation in organizations.	U	C	Instructor created exams /Assignment /Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Basic Management		11	24
	1	Introduction, Definition and characteristics of management.	1	
	2	Functions of management - Planning, Organising, Staffing, Directing, Coordination, Controlling, Motivating, Communication, Decision Making.	2	
	3	Principles of management – F. W. Taylor, Henry Fayol, Elton Mayo.	2	
	4	Administration and management, Nature of management, levels of management, managerial skills, managerial roles.	3	
	5	Forms of Organization- Line, Line–staff etc. Forms of ownerships – Partnership, Proprietorship, Joint stock, Co-operative society, Govt. Sector etc.	3	
II	Strategic Management		9	18
	6	Concept and Characteristics of strategic management –Defining strategy – Mintzberg’s 5P’s of strategy – Corporate, Business and Functional Levels of strategy.	3	
	7	Strategic Management Process.	1	

	8	Preparing an Environmental Threat and Opportunity Profile (ETOP)	2	
	9	Industry Analysis - Porter's Five Forces Model of competition. BCG Matrix – GE 9 Cell Model -Balanced Scorecard, Generic Competitive Strategies: Low cost, Differentiation, Focus.	3	
III	Quality Management		14	28
	10	Definition of quality, goalpost view of quality, continuous improvement	2	
	11	Types of quality – quality of design, conformance and performance, phases of quality management, Juran and Deming's view of quality	3	
	12	Quality Management Assistance Tools: Ishikawa diagram – Pareto Analysis – Pokka Yoke (Mistake Proofing).	3	
	13	Quality circles, Total Quality Management (TQM), Barriers to TQM, Kaizen, Five S (5S), Six sigma Quality Management Standards – concepts, methodology, principles, applications to manufacturing	3	
	14	The ISO 9001:2000 Quality Management System Standard- The ISO 14001:2004 Environmental Management System Standard- ISO 27001:2005	3	
	15	Information Security Management System	2	
IV	TQM Tools & Techniques and human resource development		14	28
	16	The seven traditional rules of quality, New management rules	2	
	17	Benchmarking, reason to bench mark, Benchmarking process, FMEA: stages and types	2	
	18	Quality function deployment, TPM- concepts, improvements needed, cost of quality, performance measures and appraisal	2	
	19	Team and team work, recognition and rewards, PDSA (Plan-Do-Study-Act) cycle, customer focus-customer orientation, customer satisfaction, customer complaints and customer retention	4	
	20	Strategic importance HRM, objectives of HRM, HR department operations, Human Resource Planning - objectives and process; human resource information system.	4	
V	Management Information Systems		12	
	Open Ended	Concept of data and information, characteristics of information, types of information. Definition of MIS, Need, Purpose and Objectives, Contemporary Approaches to MIS, Components of an information system, need to study information systems, Classification of information systems. Decision-making models, Types of decisions, Decision Support Systems. Introduction to e-		

		commerce, types – B2B, B2C, C2B, C2C etc., Business Process Re-engineering (BPR)		
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References:

1. P. Khanna, “Industrial Engineering and Management”, Dhanpatrai publications Ltd, New Delhi.
2. L.C.Jhamb , Savitri Jhamb , Industrial Management – I , Everest Publishing House .
3. Dinesh Seth and Subhash C. Rastogi, “Global Management Solutions”, Cengage Learning, Second Edition, USA.
4. B. Davis and Margrethe H. Olson, "Management Information Systems", Mc-Graw-Hill International Editions.
5. Azar Kazmi, “Strategic Management & Business Policy”, Tata McGraw Hill, New Delhi
6. Kenneth C. Laudon and Jane P. Laudon, “Management Information Systems”, 8th Edition, Pearson Education
7. K. Shridhara Bhat, “Materials and Logistics Management”, Himalaya Publishing House, Mumbai
8. M.Y. Khan and P. K. Jain, “Financial Management”, Tata McGraw Hill, New Delhi
9. Ravi M. Kishore, “Project Management”, Tata McGraw Hill, New Delhi
10. Donna C.S., Summers, “Quality Management”, 2nd Edition. Prentice Hall, Upper Saddle River, NJ. ISBN-13: 9780135005101.

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	-	-	-	-	-	2	3	-	2	-	3	-	3
CO 2	-	-	-	-	-	2	3	-	2	-	3	-	3
CO 3	-	-	-	-	-	2	3	-	2	-	3	-	3
CO 4	-	-	-	-	-	2	3	-	2	-	3	-	3
CO 5	-	-	-	-	-	2	3	-	2	-	3	-	3
CO 6	-	-	-	-	-	2	3	-	2	-	3	-	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment / Discussion / Seminar
- Internal Examination
- Practical Examination
- Final Examination (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3		✓		✓
CO 4	✓	✓		✓
CO 5		✓	✓	
CO 6	✓			✓